Full Technical Specification: Randomization, Blinding, and Withdrawal

Scope: Comprehensive Implementation Guide for Multi-Study Architecture **Target:** Development Team / System Architects

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1. Architectural Goal (الهدف المعماري)

The current system must be modified so that randomization, blinding, and withdrawal are not applied in the same way across all studies. The `study_type` must be the primary factor determining:

- Availability of Randomization, Allocation Concealment, and Blinding.
- Types of blinding available.
- Withdrawal recording methods and participant states.
- Required permissions and CRF/Statistical Analysis visibility.
- Integration with CONSORT/STROBE workflows.

Rule: Do not create separate applications for each study type. All types must exist within the same platform using a **Study-Type Configuration Engine**.

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2. Supported Study Types

The system must support the following types centrally:

```
type StudyType = "RCT" | "PROSPECTIVE" | "RETROSPECTIVE" | "CROSS_SECTIONA
```

Each type must have an independent configuration.

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3. Randomized Controlled Trial (RCT)

3.1 Randomization

Randomization is allowed and enabled. The system must support:

- Simple Randomization
- Permuted Block Randomization (Fixed & Variable)
- Stratified Randomization (and Stratified Block)
- *Scalability:* The system should be extensible to support Cluster, Minimization, and Adaptive Randomization in the future.

3.2 Randomization Sequence

- **Security:** DO NOT use `Math.random()` for production. Use a **Cryptographically Secure Random Number Generator**.
- **Immutability:** The sequence is immutable after activation. Any modification requires a new version with a full Audit Trail.
- **Stored Attributes:** `study_id`, `sequence_id`, `method`, `block_size`, `stratification_factors`, `seed/version`, `generated_at`, `generated_by`, `status`.

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4. Allocation Concealment (RCT)

Randomization must be decoupled from Allocation Concealment. Supported methods:

- Central Web Randomization
- Sequentially Numbered Opaque Sealed Envelopes (SNOSE)
- Central Allocation / Pharmacy-controlled / Independent Officer.

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5. Blinding in RCT

Supported Models: **Open Label, Single Blind, Double Blind, Triple Blind**. It is not enough to store `is_blinded = true`. You must define the blinded roles:

- Participant, Researcher, Outcome Assessor, Data Entry Staff, Statistician, Supervisor.

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6. Role-Based Blinding (Server-Side)

Blinding must be enforced on the **Server-side**, not just hidden in React.

- **Example:** An Outcome Assessor might see Participant ID and Visit data but MUST NOT see the Treatment Group or Actual Intervention, even if they access the API directly.
- **Enforcement Layers:** Frontend, Backend, API, Database Access, Authorization Layer.

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7. Treatment Codes

Decouple **Actual Treatment** from **Masked Treatment Code**.

- *Actual:* “Orthodontic Intervention A”
- *Masked:* “ARM_A” Only authorized personnel should see the actual intervention; others see the masked code.

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8. Emergency Unblinding

Mandatory workflow for RCT:

1. **Request:** Participant ID + Reason.
2. **Justification:** Clinical Justification.
3. **Authorization:** Approval process.
4. **Reveal:** Allocation revealed for that specific participant only.
5. **Audit:** Log all details (who, when, why, what was revealed).

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9. Final Unblinding

Different from Emergency Unblinding. Occurs only after:

- Recruitment & Follow-up Complete → Data Cleaning → Data Freeze → Database Lock → Blinded Statistical Analysis → Approval → **Final Unblinding**.

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10. Statistician Blinding

If blinded, the statistician sees `ARM_A / ARM_B` instead of `Experimental / Control`. Analysis continues this way until Database Lock and Blinded Analysis approval.

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11. RCT Withdrawal States

“Withdrawn” should not be a single state. Use granular states:

- Active, Completed, Lost to Follow-up.
- Withdrawn (Participant Decision, Investigator Decision, Safety, Protocol Violation).
- Discontinued Intervention, Death, Screen Failure, Randomized but Never Treated.
- **History:** Maintain a timestamped history of all state transitions.

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12. Data Retention Policy

Withdrawal ≠ Deletion. When a participant withdraws, do not delete their record or CRF data.

- Set `status = WITHDRAWN`.
- Store `withdrawal_date`, `reason`, `category`, `requested_by`, `last_visit`, `follow_up_status`.

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13. Prospective Study

- **Randomization:** OFF by default. Can be enabled if the design requires it (with a warning).
- **Blinding:** Optional (e.g., for Outcome Assessors).
- **Allocation Concealment:** Only visible if an intervention/allocation process is active.
- **Withdrawal:** Similar to RCT but without randomization records unless enabled.

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14. Retrospective Study

- **Randomization/Allocation/Blinding:** DISABLED.
- **UI:** Hide these modules entirely.
- **Focus:** Data Source (Existing Records), Data Masking (De-identification/Pseudonymization).
- **Note:** Data Masking is not Randomization.

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15. Retrospective Withdrawal

- **States:** Record Included, Excluded, Withdrawn from Dataset, Data Unavailable, Insufficient Records, Duplicate, Ineligible.
- **Action:** Log `data_withdrawal_date`, reason, and authorized person. Do not delete Audit Trail.

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16. Cross-Sectional Study

- **Randomization/Allocation:** DISABLED.
- **Blinding:** Optional for Assessors/Statisticians to prevent bias.
- **Withdrawal:** Invited, Screened, Eligible, Enrolled, Assessment Completed, Withdrawn, Excluded, Lost/Incomplete.

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17. Study-Type Capability Matrix

Feature	RCT	Prospective	Retrospective	Cross-sectional
Randomization	YES	OPTIONAL	NO	NO
Allocation Concealment	YES	CONDITIONAL	NO	NO
Blinding	YES	OPTIONAL	OPTIONAL	OPTIONAL
Emergency Unblinding	YES	CONDITIONAL	NO	NO
Final Unblinding	YES	CONDITIONAL	NO	NO
Statistician Masking	YES	OPTIONAL	YES	OPTIONAL
Withdrawal	YES	YES	DATA WITHDRAWAL	YES
Lost to Follow-up	YES	YES	NO/NA	OPTIONAL

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18. StudyCapabilityEngine (Backend Authority)

Do not rely solely on Frontend `if` statements.

```
interface StudyCapabilities {
  randomization: boolean;
  allocationConcealment: boolean;
  blinding: boolean;
  emergencyUnblinding: boolean;
  finalUnblinding: boolean;
  statisticianMasking: boolean;
  withdrawal: boolean;
  lossToFollowUp: boolean;
}
```

The backend must validate these capabilities before any operation.

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19. Participant State Machine

Do not store status as a simple string. Create a `participant_status_history` table.

- **Fields:** `from_status`, `to_status`, `changed_by`, `changed_at`, `reason`.

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20. Database Structure (Minimal Requirements)

Review and extend existing schema to include:

- `studies`, `participants`, `participant_status_history`.
- `randomization_sequences`, `randomization_assignments`, `allocation_records`.
- `blinding_configurations`, `blind_role_permissions`, `treatment_codes`.
- `unblinding_requests`, `unblinding_events`, `withdrawal_records`, `audit_logs`.

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21. Security & Permissions

- **Actual Treatment Allocation:** Highly sensitive. Must not be accessible to all users.
- **Requirements:** RBAC + Study-level permissions + Server-side filtering + Audit Trail.

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22. Audit Trail Rules

Log all sensitive operations: Randomization generation, Allocation reveal, Unblinding (Request/Approval/Reject/Emergency), Withdrawal, Status Change, Database Lock.

- **Attributes:** `user_id`, `study_id`, `participant_id`, `action`, `timestamp`, `old_value`, `new_value`, `reason`.
- **Note:** Users cannot delete Audit Logs.

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23. UI Behavior per Study Type

- **RCT:** Shows all modules (Randomization, Blinding, etc.).
- **PROSPECTIVE:** Shows Randomization/Blinding as Optional.
- **RETROSPECTIVE:** Shows Data Masking/Withdrawal; Hides Randomization/Allocation/Emergency Unblinding.
- **CROSS_SECTIONAL:** Shows Optional Assessor Blinding/Statistician Masking.

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24. Integration (CRF & Statistics)

- **CRF:** Filter data based on role (e.g., hide allocation from blinded Assessors).
- **Statistics:** Use masked datasets (`ARM_A/B`) until Final Unblinding. Support ITT, Per Protocol, and Safety populations based on SAP.

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25. Acceptance Criteria

The implementation is complete only if:

- **RCT:** Randomization is immutable; Blinding is enforced server-side; Emergency unblinding is audited; Withdrawal preserves data.

- **Prospective:** Randomization is optional; Withdrawal workflow is functional.
- **Retrospective:** Randomization is disabled; Data masking/withdrawal works; Audit trail is preserved.
- **Cross-sectional:** Optional masking; Withdrawal workflow works.

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26. Critical Implementation Rule

DO NOT rebuild the Clinical Research Platform. Inspect existing components, identify reusable parts, and integrate the new logic into the existing codebase.